

Why Important Questions Are Often Difficult

Significance, Persistence, and the Composition of Unresolved Inquiry

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Abstract

Why do questions that matter so much often remain difficult to answer? This paper distinguishes obstacles to resolving a particular question from the processes that shape which questions remain under active inquiry. Questions may stay open because evidence, methods, or other necessary conditions are missing. Across a population, difficulty can slow resolution while significance can sustain pursuit. As more tractable questions are answered and less-valued inquiries are dropped, the remaining active questions can become richer in both significance and difficulty. A competing-risks model shows that this enrichment does not necessarily create statistical dependence: with independent attributes at entry and separable survival weights, the two properties remain independent among survivors. Examples from disease research, mathematics, markets, and artificial intelligence distinguish persistent obstacles from changing questions. The practical task is to identify why a particular inquiry remains open before deciding what further progress requires.

Keywords: research questions; significance; objective difficulty; competing risks; research persistence; disengagement; scientific problem choice

1. Introduction

Some of the questions that matter most to us are also among the hardest to settle. Serious diseases can remain subjects of inquiry for decades. A mathematical conjecture can withstand generations of attention. A profitable opportunity can change when others pursue it, and a technological advance can answer yesterday's question while creating tomorrow's. These cases raise a deceptively simple question: does importance make questions difficult, or are we seeing a selected set of questions that have stayed open?

There are at least two parts to the answer. A particular question may be hard to resolve because a necessary observation, method, instrument, or intermediate result is unavailable. Separately, the population of questions still being pursued changes over time. Easier questions can leave that population when answered; less-valued questions can leave when pursuit stops. Questions that are difficult yet still consequential may remain visible through both processes. This does not imply that significance itself creates difficulty.

To understand the population effect, distinguish two exits from active unresolved inquiry. A question leaves when it receives an adequate answer or when meaningful pursuit ends. Difficulty can slow resolution. Significance can sustain pursuit when resources and institutions permit. The two exits therefore select questions for different reasons.

Imagine two equally difficult questions. One matters greatly to the people studying it; the other offers little value. The first may attract sustained effort while the second loses support. Now imagine two equally consequential questions, one tractable and one difficult. The tractable one may be answered sooner. Repeated across a population, these decisions can leave an active set unusually rich in important, difficult questions—without having made any individual question harder.

We begin by defining the questions being followed and distinguishing significance, difficulty, and unresolved status. We then examine the two exits and the conditions that can obstruct resolution, using four contrasting domains to show why inquiry histories differ. Related research places these processes in context. Finally, a competing-risks calculation establishes what population selection can explain—and why enrichment in both attributes does not necessarily make them statistically associated.

2. Problems, Questions, and the Unit of Inquiry

Problems and questions are related but not identical. A problem is a broader condition, challenge, opportunity, or domain of concern. It can generate many questions, some already settled and others still open. The questions are the units of inquiry followed in this paper.

A broader problem can remain consequential after particular questions about it have been answered. Chronic kidney disease, for example, continues to motivate research alongside established knowledge about its causes, diagnosis, and management [1]. Conversely, a specific question may have substantial scientific value while addressing only a limited part of a practical problem. Treating the entire domain as one question would conceal which inquiries have been resolved and which remain open.

We therefore follow a defined question within a specified inquiry community or program. The question includes the target and the standard for an adequate answer. Its significance, difficulty, and unresolved status can then be distinguished without confusing the condition that motivates inquiry with the inquiry itself.

2.1 Question significance

Significance is the consequence or value of resolving the defined question, assessed for specified purposes and stakeholders. It may concern a decision, scientific understanding, a practical outcome, or management of the broader problem. The same question can have different significance for different

communities. A population comparison therefore needs a stated assessment perspective; the importance of the underlying problem alone does not establish the value of resolving each question it generates.

2.2 Objective question difficulty

Objective difficulty is resistance to adequate resolution under the available inquiry conditions. Evidence, methods, tools, expertise, coordination, and target stability can each affect that resistance. We use objective to direct attention to those conditions, rather than an investigator's frustration or the mere fact that no answer has yet been found. Difficulty remains relative to the current inquiry environment: a new instrument or method can make the same question more tractable.

2.3 Unresolved status

Unresolved status records that an adequate answer has not been reached. It does not establish why. A question may resist sustained competent inquiry, receive little serious attention, or lack an identifiable condition needed for the chosen inquiry strategy. Undone-science research documents socially valuable questions left incomplete or unstudied as funding, institutions, and power shape research agendas [2]. We therefore need the question's pursuit history and inquiry conditions before interpreting its non-resolution as evidence of difficulty.

3. The Population of Active Unresolved Questions

Having distinguished those properties, we can identify the population to be followed. Some possible questions never receive sustained attention; others enter active inquiry within a defined community or program. We call a question active when meaningful pursuit continues under an observable, domain-appropriate criterion. Withdrawal by one investigator counts as disengagement only if that criterion is no longer met for the unit being tracked.

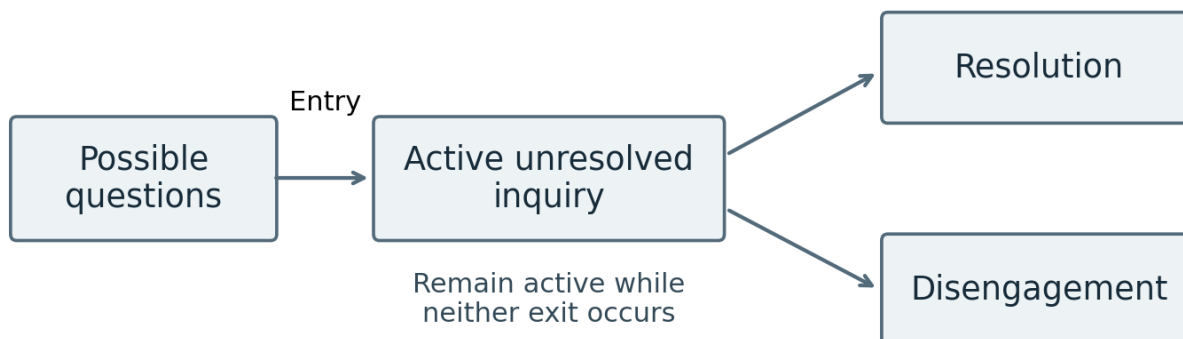
Once active, a question may reach an adequate answer or leave active pursuit while still unresolved. Dormancy and revival create further episodes, and a changed target can require recording a new question rather than treating it as the same inquiry. Resolution can also expose successor questions. Sequential scientific problem-choice models already represent such renewal, including Carayol and Dalle's account of open science [3].

Figure 1 separates entry from the two exits. Questions that have reached neither exit by a specified time remain in the active unresolved state. They form a survivor population, selected from the questions that entered inquiry; the exits do not lead to that population, but remove questions from it.

The selection reflects different histories. Tractable questions may leave when answered, while questions judged no longer worth pursuing may leave through disengagement. The remaining population reflects both processes, together with the conditions that shaped entry. A cross-section of active unresolved questions therefore cannot reveal those histories by itself.

Resolution and disengagement convey different information. Resolution indicates that inquiry reached an adequate answer under the relevant standard. Disengagement can reflect value, cost, institutional conditions, or changing priorities.

Figure 1. Entry, Active Inquiry, and Two Principal Exits



3.1 Question difficulty and population difficulty

This population distinction helps interpret progress. A field can answer many questions while its remaining active questions are demanding, because resolved questions have left the set. Progress can also open deeper questions or change an adaptive target. Bloom and colleagues document settings in which research effort rises while research productivity falls [4]. That evidence concerns productivity, rather than a direct test of our question-level exit account, but it reinforces the need to distinguish successful work from the characteristics of the work still ahead.

4. Two Exit Processes

Two tendencies can shape the active unresolved population. Under comparable inquiry conditions, greater difficulty can slow an adequate answer. Greater significance can sustain effort, reducing disengagement when continued pursuit is possible. The first tendency retains questions that resist resolution; the second retains questions that remain worth pursuing. Table 1 describes the possible combinations, not an inevitable outcome for every question.

Table 1. Significance, Difficulty, and Typical Exit Pressure

Significance	More tractable	More difficult
Lower	Can leave through resolution or disengagement when pursuit loses priority.	Can remain technically open while active pursuit ends.

Significance	More tractable	More difficult
Higher	Continued attention can bring an answer relatively quickly.	Can remain active while a consequential question resists resolution.

These tendencies can increase the **shares** of high-significance and high-difficulty questions among survivors even as the absolute number of active questions falls. Whether the two attributes also become statistically associated is a separate question, addressed in Section 8.

The influences need not remain confined to separate exits. Significance can attract resources that help resolve a question. Difficulty can make pursuit prestigious or discourage further effort when the prospect of success seems poor. The resulting population depends on how these additional influences operate.

5. Resolution Constraints

The population account explains why certain questions remain visible; it does not, on its own, explain why any one question resists an answer. For that, we need to identify obstacles to resolution independently of the outcome. A question is not difficult simply because it has remained unanswered.

For a causal inquiry, an available dataset may not contain the contrast needed to distinguish competing explanations. For a mathematical proof, a proposed route may depend on an intermediate result that has not been established. Either constraint is specific enough to investigate, and a different method may overcome it. Other limitations can come from computation, instrumentation, expertise, coordination, or a changing target. What matters is identifying which condition prevents the proposed route from delivering an adequate answer.

This distinction links individual difficulty to population selection. We can study whether independently assessed obstacles slow resolution, then ask how the resulting exits change the mix of questions still pursued. The following examples illustrate why those two levels should be considered together.

6. Selected Dynamics

6.1 Disease

Chronic kidney disease illustrates the difference between an enduring problem and the questions it generates. It remains a serious health concern, although established knowledge identifies diabetes and high blood pressure as common causes in adults and supports approaches to

diagnosis and management [1]. The broader problem therefore persists even as some questions about it have been answered.

Imagine two further inquiries within such a field: one asks whether an established diagnostic procedure identifies a specified condition under known circumstances; another asks which of several poorly observed mechanisms explains a particular progression pattern. The first may be answerable with available evidence; the second may require observations or methods that do not yet exist. These are illustrative questions, not claims about a particular current clinical research agenda. The lesson is that the importance of the disease cannot, by itself, tell us which questions are unresolved or why.

6.2 Longstanding mathematics

The Riemann hypothesis offers a contrasting case: a defined mathematical conjecture whose target has remained stable since its formulation in 1859. It is still listed as an unsolved Millennium Prize Problem [5]. Its long history of serious attention makes persistence visible, but the history alone cannot identify the mathematical obstacle responsible for the lack of an accepted proof or disproof.

Here the question itself does not move when researchers change their methods. A successful new approach could resolve it; continued work can also clarify why familiar approaches fail. The example shows why significance can sustain inquiry on a stable target while explaining difficulty still requires attention to the technical work.

6.3 Markets

Markets offer a different kind of challenge: acting on an answer can change the conditions under which it holds. Consider an initially profitable opportunity. An investor may learn that it can generate excess returns, but imitation and competition can narrow the opportunity before the same answer is useful again. Rosenbaum's industry-level evidence that profits attract entry, subject to entry barriers [6], supports the possibility of competitive response; it does not directly measure the life of an investment strategy.

This is an illustrative adaptive mechanism rather than a documented sequence for a particular strategy. The inquiry may need to ask not merely whether an opportunity worked, but whether it remains exploitable under current conditions. In such settings a question can become difficult because the target changes, not only because the investigator lacks a method for a fixed target.

6.4 Artificial intelligence

Artificial intelligence provides another familiar progression. A benchmark that once separated stronger from weaker systems can become less informative as many systems achieve similarly high scores. The initial comparison may have been answered, but a new question then arises: what evaluation distinguishes the capabilities that now matter? Reports of changing model performance and evaluation practices provide context for this pattern [7].

Improved models and methods can therefore reduce the difficulty of one fixed question while making a successor question more urgent. In governance, the same pattern may appear when a newly practical use changes which consequences need to be assessed. This is renewal of the inquiry population, not evidence that every individual AI question grows harder.

6.5 What the selected dynamics show

The four cases show why the same label—unresolved—can hide different histories. A disease domain contains both answered and unanswered questions; a mathematical conjecture offers a stable target; competitive markets can change the target; and AI advances can generate new questions as old ones become tractable. Before comparing their difficulty, we need to know which question, which conditions, and which episode of inquiry we are following.

7. Related Literature

7.1 Problem choice and search

The first question concerns entry: why are some possible inquiries pursued? Carayol and Dalle [3] model sequential scientific problem choice and successor generation within an evolving knowledge structure. Problemistic-search research examines organizational search prompted by performance below aspirations [8]. Rzhetsky and colleagues [9] represent importance and difficulty in a model of experiment choice, while Alon [10] discusses interest and feasibility in scientific problem selection. These approaches show why the population at entry already reflects choice; independence between significance and difficulty at entry must be an explicit assumption or an empirical finding.

7.2 Persistence, disengagement, and abandonment

A second literature concerns whether pursuit continues. Psychology distinguishes persistence from disengagement in goal pursuit [11]. Undone-science research shows that socially important questions can remain incomplete because research agendas are institutionally structured [2]. Operations research has modeled when an uncertain research project should be abandoned and replaced by another opportunity [12]. Innovation research likewise studies continuation, delay, abandonment, and completion within project portfolios [13], [14].

These strands make disengagement a substantive research object rather than a residual category. They also show that leaving an unresolved question can be rational, institutional, or resource-driven.

7.3 Selection and population composition

After entry and exit, the surviving set is a selected population. Research on endogenous selection explains how conditioning on selection can alter observed relationships among variables [15]. R&D portfolio research examines ranking, continuation, cancellation, and project selection [16]. We apply the population logic to unresolved questions, but leave open whether the selection changes the joint relationship between significance and difficulty. As the formal analysis will show, selection can change both marginal distributions while preserving their independence.

7.4 Changing populations of unresolved questions

A further literature explains why progress need not produce an easier remaining set. Bloom et al. [4] document settings in which research effort rises while research productivity falls. Sequential problem-choice models allow resolved work to generate successor questions [3]. Adaptive domains add another channel: inquiry or exploitation can change the target itself.

Together, these literatures help explain which questions enter inquiry, which continue to receive effort, which exit, and how successor questions arise. The remaining analytical task is to determine what those flows imply for the composition of an active cohort.

8. Formal Representation

8.1 The cohort and the inquiry episode

We now formalize one part of the story: how two exits alter a defined cohort of questions. A question leaves the active unresolved state through resolution or disengagement; the first such event ends the episode. Cause-specific hazards describe the rates of these exits among questions still active [17]. This framework does not require independent latent times to the two exits.

Let $i = 1, \dots, n$ index questions entering a cohort at elapsed time $t = 0$. Each question has a defined target and adequacy standard. Let T_i be its elapsed time to first exit, and let $J_i \in \{R, A\}$ identify resolution (R) or disengagement (A). Time is measured in one consistent unit, such as years after entry. Questions with $T_i > t$ remain active and unresolved at time t .

Let S_i and D_i denote assessed significance and difficulty. Higher values indicate greater significance and greater resistance to resolution, respectively. We hold those attributes fixed within this baseline calculation and compare questions under a common observed inquiry profile w , such as resources and institutional conditions. The scales are domain-specific, and neither attribute is defined by its later exit

time. This restriction isolates selection from changes in the attributes themselves; changing environments and revived episodes require the extension discussed in Section 8.5.

8.2 The rates of the two exits

For exit type $k \in \{R, A\}$, define its cause-specific hazard among questions that have not yet exited:

$$\lambda_k(t \mid s, d, w) = \lim_{h \downarrow 0} \left(\frac{\Pr(t \leq T_i < t+h, J_i=k \mid T_i \geq t, S_i=s, D_i=d, w)}{h} \right). \quad (1)$$

Here h is a short positive time interval, and s and d are values of the two attributes. The hazard has units of inverse time. It is an instantaneous rate among survivors, rather than the probability of an eventual resolution or disengagement. We assume continuous exit times and nonnegative, integrable hazards over the intervals considered.

To represent the proposed directional tendencies, let $\boldsymbol{\eta}_R$ and $\boldsymbol{\eta}_A$ be coefficient vectors for the common profile w . A possible proportional-hazards specification is

$$\lambda_R(t \mid s, d, w) = \lambda_{R0}(t) \exp\{-\beta_D d + \theta_S s + \boldsymbol{\eta}_R^\top w\}, \quad (2)$$

$$\lambda_A(t \mid s, d, w) = \lambda_{A0}(t) \exp\{-\gamma_S s + \kappa_D d + \boldsymbol{\eta}_A^\top w\}. \quad (3)$$

The baseline hazards $\lambda_{R0}(t)$ and $\lambda_{A0}(t)$ supply the time-dependent rates at zero-valued predictors. Superscript $\top$ denotes transposition, so each vector product is a sum of coefficient-weighted profile components. All terms in each exponent are dimensionless; coefficients are interpreted relative to the units of their corresponding assessment scales.

Holding the other predictors fixed, $\beta_D > 0$ represents a lower resolution rate at greater difficulty, and $\gamma_S > 0$ represents a lower disengagement rate at greater significance. The cross-effects θ_S and κ_D allow significance to affect resolution and difficulty to affect disengagement. For example, significance may mobilize resources, while difficulty may discourage continued pursuit. These equations are proposed specifications, not estimated causal findings; their signs and adequacy require empirical assessment.

8.3 From exit rates to the surviving population

We next connect those rates to population composition. For clarity, hold w common and suppress it in the notation. The total rate of leaving the active unresolved state is the sum of the two exit rates:

$$\lambda(t \mid s, d) = \lambda_R(t \mid s, d) + \lambda_A(t \mid s, d). \quad (4)$$

Let $G(t \mid s, d)$ be the probability of remaining in that state to time t . Over a short interval, the fraction leaving is approximately the total hazard multiplied by the interval length. This gives $\partial G / \partial t = -\lambda(t \mid s, d)G(t \mid s, d)$, with $G(0 \mid s, d) = 1$. Integrating yields

$$G(t \mid s, d) = \exp \left\{ - \int_0^t [\lambda_R(u \mid s, d) + \lambda_A(u \mid s, d)] du \right\}. \quad (5)$$

The variable u denotes elapsed time within the integral. A greater accumulated total exit rate produces a smaller survival probability. Because the rates are nonnegative, G lies between zero and one and cannot increase within this first-exit episode. At entry, every cohort member is active, so the survival probability is one.

Let $f_0(s, d)$ be the joint density of significance and difficulty at entry within the common profile. Survival multiplies each entry type by its probability of remaining. Dividing by the overall surviving fraction gives the density among survivors:

$$f_t(s, d) = \frac{f_0(s, d)G(t|s, d)}{\iint f_0(a, b)G(t|a, b) da db}. \quad (6)$$

The dummy variables a and b range over the significance and difficulty support. The denominator must be positive for a survivor distribution to exist. For discrete categories, sums replace the integrals. At $t = 0$, the expression recovers the entry distribution f_0 . Later, types that remain longer acquire greater relative weight, even though their individual survival probabilities and expected counts can only fall.

8.4 Enrichment and independence

Begin with 100 questions, 25 in each combination of higher or lower significance and greater or lesser difficulty. At a selected time, suppose easier questions retain a survival factor of 0.4 and harder questions 0.8, while lower-significance questions retain a factor of 0.4 and higher-significance questions 0.8. Multiplying the two factors gives the expected counts in Table 2.

Table 2. Entry counts and expected survivors under separable weights

Significance	Entry: easier	Entry: harder	Expected survivors: easier	Expected survivors: harder
Lower	25	25	4	8
Higher	25	25	8	16

$$16/36 = 4/9$$

$$25/100 = 1/4$$

The example leaves 36 expected survivors. Harder questions and higher-significance questions each make up two-thirds of that group, compared with half at entry. The high-significance, high-difficulty combination accounts for 16 of 36 survivors rather than 25 of 100 entrants. Its share has risen even though its expected count has fallen. Actual counts in a finite sample could differ from these expectations.

Here is the surprising point: enrichment has not made significance and difficulty statistically dependent. Among survivors, two-thirds are harder questions at either significance level.

Knowing a surviving question's significance therefore tells us nothing further about which difficulty category it occupies. The algebra below shows why.

To show why, consider the sufficient special case in which difficulty alone affects resolution and significance alone affects disengagement, within the common inquiry profile:

$$\lambda_R(t | s, d) = r(t, d), \quad \lambda_A(t | s, d) = a(t, s). \quad (7)$$

Define the two survival factors by their accumulated rates:

$$G_D(t | d) = \exp \left\{ - \int_0^t r(u, d) du \right\}, \quad G_S(t | s) = \exp \left\{ - \int_0^t a(u, s) du \right\}. \quad (8)$$

Substituting the separate rates into the survival expression and splitting the exponential gives

$$G(t | s, d) = G_D(t | d)G_S(t | s). \quad (9)$$

The survivor weight thus separates into one difficulty-related factor and one significance-related factor. This factorization, combined with independence at entry, is what preserves independence among survivors.

$$f_0(s, d) = f_{0S}(s)f_{0D}(d). \quad (10)$$

Here f_{0S} and f_{0D} are the entry marginal densities. In the survivor-density expression, both the numerator and the normalizing denominator now separate into the same two components:

$$f_t(s, d) = \frac{f_{0S}(s)G_S(t|s)}{\int f_{0S}(a)G_S(t|a) da} \frac{f_{0D}(d)G_D(t|d)}{\int f_{0D}(b)G_D(t|b) db}. \quad (11)$$

Each factor is a normalized marginal survivor density. Their product establishes independence among survivors. If harder questions have lower resolution rates and more significant questions have lower disengagement rates, each marginal shifts toward larger values, while the joint distribution still factorizes. The calculation therefore establishes enrichment without dependence.

Assigning each attribute to a different exit is one sufficient way to obtain factorization, not the only way. Cross-effects can exist without creating dependence if the *total* survivor weight still factorizes. Conversely, dependence already present at entry, non-factorizing selection, or mixing unlike cohorts can yield association. The mathematical result is about a specified survivor population, not a universal law tying importance to difficulty.

8.5 What empirical work would need

The calculation identifies distinct empirical tasks. A study needs a defined cohort, a common observation horizon or an explicit treatment of cohort age, and independently assessed significance and difficulty. It must then observe entry, resolution, disengagement, and relevant inquiry conditions. Those records allow tests of whether difficulty slows resolution, significance sustains pursuit, and non-separable selection or prior dependence contributes to an observed association.

When difficulty or significance changes during inquiry, the fixed-attribute calculation is insufficient. The hazards can condition on measured histories of those attributes and inquiry conditions; survival then depends on the path taken. The distribution of current attributes requires tracking those changes rather than simply reweighting the entry density. Likewise, revival, successor generation, and continued entry require new episodes or an expanded population model.

Biomedical research portfolios, mathematical open-question records, AI benchmark histories, and R&D portfolios offer possible settings. Each requires a defensible mapping from recorded projects or benchmark results to defined questions and adequacy standards. Observation ending while a question remains active should also be distinguished from disengagement. These design choices determine which part of the population argument the evidence can actually test.

9. Implications

The analysis returns us to a practical question: why is this consequential inquiry still open? A missing instrument calls for capability development; a weak evidence base calls for different observations; limited attention may call for a resource decision; and a moving target calls for a revised question or standard of adequacy. The appropriate response depends on the history, not merely on the question's unresolved status.

A research portfolio requires the same distinction. Many demanding questions can remain because tractable questions were successfully answered and hard but valuable questions continued to receive support. Yet important tractable questions may also be missing because they never entered inquiry or were abandoned. Looking only at the active unresolved list cannot distinguish achievement from neglect.

This is why a field can make substantial progress without appearing to have an easier set of remaining questions. Understanding progress means recording what was answered, what was dropped, and what new questions entered—not simply counting what remains open.

10. Limitations

The model follows a defined cohort through its first exit from active unresolved inquiry. Actual research communities continuously admit new questions, change their priorities, and sometimes reopen older inquiries. Those movements require a broader population or multi-state representation; they cannot be inferred from the baseline survivor formula alone.

Empirical application also needs credible, domain-specific assessments of significance and difficulty. Their measures must not be defined by the same resolution outcomes whose effects

are being tested. Differences in stakeholders, resources, and adequacy standards further limit comparisons across fields.

Finally, apparent inactivity is not always disengagement. Low-intensity pursuit, dormancy, revival, changed targets, and contested resolutions need observable treatment. The examples in this paper identify possible mechanisms; they do not estimate their prevalence or causal magnitude.

A useful empirical test would therefore track entries and exits, assess constraints independently, and distinguish changing questions from stable ones. These requirements mark the boundary between the analytical explanation offered here and claims about how frequently it operates in any particular field.

11. Conclusion

Why are important questions so often difficult? Part of the answer concerns the questions themselves: some cannot be adequately resolved with the evidence, methods, or other conditions currently available. Another part concerns what remains in view: tractable questions leave when answered, while consequential questions may continue to attract effort despite their difficulty.

The two exits can make the surviving population richer in both significance and difficulty without making either attribute cause the other. The formal result shows why such enrichment is compatible with statistical independence; an observed association requires additional scrutiny of entry conditions and survivor selection.

The picture also changes as inquiry progresses. New instruments and methods may open routes to a fixed answer. Successful work can reveal deeper questions. In adaptive settings, the target itself can move. An unresolved population can therefore remain demanding even while meaningful progress continues.

What remains visible is shaped by what we have answered, what we have stopped pursuing, and what still matters enough to keep trying. Before deciding whether to invest further effort, we should recover the particular question's history and identify what currently prevents an adequate answer. Persistence is a reason to investigate those conditions—not an explanation in itself.

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